

**The Reachable Human Phenotype:
Factorized Selfhood, Morphology-Congruent Binding, and
What the Decisive Experiment Costs**

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Abstract

Cognitive neuroscience studies capability, body ownership, agency, and identity in separate literatures, and each has independently shown that its object moves. Synchronous multisensory evidence gives people ownership of rubber hands, third arms, and virtual cat ears; avatars feed back into behavior; and communities exist whose members report persistent nonhuman self-models or cultivated inner agents. This paper treats those findings as coordinates of one object, the reachable human phenotype, defined as the set of configurations a nervous system can enter through physiological change, sensory perturbation, learning, and context. Ten coordinates are proposed and the factorization is forced rather than tidy: a 2026 experiment raised subjective ownership of virtual cat ears in 24 participants while proprioceptive drift did not follow, so ownership and body schema moved apart in one study, and any scalar strength-of-alternate-self would have erased the field's cleanest dissociation. The paper's hinge is an awkward empirical result. The one direct study of body ownership in a nonhuman-identifying community found 50 furries reporting weaker embodiment of a human rubber hand, the opposite sign from a global-plasticity account. A causal-inference model of ownership with a morphology-congruence prior reproduces that sign and predicts stronger congruent-nonhuman ownership from the same shifted parameter, moving 0.22 in each direction: one knob, two signs, so the anomaly becomes support. The central practical contribution is a design analysis. Simulating three live theories and fitting all three back, the design the literature has run cannot tell them apart, recovering the truth at 0.51 and misreading morphology-congruent binding as global plasticity 96 percent of the time; adding a generic nonhuman avatar reaches 0.73; only a crossed design manipulating morphology against identity congruence reaches 0.94, and it needs about 30 participants per group for 80 percent power on the interaction rather than the 20 typical of embodiment work. A multiscale formalism then separates fast neural state from slowly learned model parameters, makes hysteresis measurable as the 55 percent of a trained configuration that survives after the perturbation stops, and states the SELF and OTHER binding hypothesis as a crossed prediction rather than a main effect. Nothing here is evidence about any person or community; the models exist to say which experiment would be, and what it would cost.

The default and the reachable

An ordinary waking adult has one body, one set of capabilities, one apparent author of their thoughts. Each of those looks like a fact about the organism, and each has been moved in a laboratory. Ownership of a rubber hand follows synchronous visuotactile stimulation (Botvinick and Cohen, 1998). Ownership extends to a third arm that nobody has (Guterstam, Petkova and Ehrsson, 2011), to supernumerary robotic limbs whose incorporation shows up in peripersonal-space measures rather than questionnaires alone (Arai et al., 2022), and to avatars with nonhuman morphology that participants learn to control (Won, Bailenson, Lee and Lanier, 2015; Krekhov, Cmentowski and Krüger, 2019). The avatar then feeds back: people given taller or more attractive bodies behave differently afterward (Yee and Bailenson, 2007).

If the phenotype moves along that many axes, the ordinary configuration is a point in a space rather

than the shape of the organism. This paper calls that space the reachable human phenotype: the set of behavioral and phenomenological configurations a nervous system can enter through physiological change, sensory perturbation, learning, and context. The question is what its coordinates are, and what it would cost to find out which of them move independently. The paper's answer to the last question is a number, and the number is the contribution most likely to be useful.

Ten coordinates, and why one number will not do

The proposed coordinates are capability, body ownership, body schema, self-location, agency, identity, memory access, perceptual organization, affect, and reality testing. The precedent for factorizing at all is established: bodily self-consciousness already decomposes into ownership, self-location, and first-person perspective, which dissociate under different manipulations (Blanke, Slater and Serino, 2015), and the self that does the owning is itself inferential, built partly from interoceptive prediction (Seth, 2013).

The strongest single reason to keep the coordinates apart is a 2026 experiment. Twenty-four participants were given virtual cat ears with congruent visual, tactile, and auditory manipulation; subjective ownership of the ears rose and referral of touch and sound followed, while proprioceptive drift did not move (Yamamura, Kondo, Sakurada and Normand, 2026). Ownership changed without evidence that the ears had become a spatially grounded part of the body schema. A companion result the same year found that active and passive haptics increase the sense of embodiment for virtual dog ears in 28 participants (Khan, Trinh, Lisle and Do, 2026). Had the field been measuring one scalar called strength of alternate embodiment, the cat-ear dissociation would have been invisible, and it is the cleanest structural result available. The methodological warning is already in the literature: a meta-analysis across 111 articles finds modest ownership effects, somewhat stronger agency effects, and pervasive problems with power and inconsistent terminology (Mottelson, Muresan, Hornbæk and Makransky, 2023).

What the record establishes, and what it does not

Three communities have become relevant to this question, and the discipline that matters most is stating exactly what is known. Therians identify as or with a nonhuman animal; furry fandom involves anthropomorphic animal characters, with a fursona often functioning as a self-representation; tulpamancers deliberately cultivate an inner agent experienced as having its own perspective. These are communities with their own vocabularies rather than clinical categories, and nothing in this paper infers disorder from unusual content.

The quantitative record is thin and real. In 112 therians against 265 comparison participants, unusual experiences, introvertive anhedonia, and autism-quotient scores predicted therian status, with the odds of exceeding the autism-quotient cutoff about 5.9 times higher and a wide interval running from 2.04 to 17.31 (Clegg, Collings and Roxburgh, 2019). The design is cross-sectional. The causal direction is unresolved. Phenomenological work documents phantom body parts and shifts in 5 in-depth interviews (Grivell, Clegg and Roxburgh, 2014), and a 2025 phenomenological

study finds these experiences involving body, perception, environment, affect, and behavior rather than belief alone (Luiggi-Hernández, Fein, Bradley and Pelly, 2025). Liking an animal and identifying as one are psychometrically separable, which a self-binding account requires (Roberts, Plante, Gerbasi and Reysen, 2015). On the agent side, 243 participants across tulpamancy, ASMR, and comparison groups showed unusual sensory experience without a general elevation in delusion-proneness (Palmer-Cooper, McGuire and Wright, 2021), and experienced character autonomy is not confined to any community, since adult fiction writers report it too (Taylor, Hodges and Kóhanyi, 2003). Clinical therianthropy exists as a separate phenomenon with its own review, whose authors caution explicitly against equating nonclinical identities with clinical presentations, and this paper cites it strictly as a boundary condition (Blom and Sharpless, 2025).

What does not exist is any study comparing these groups under matched embodiment and agency manipulations, and no therian-specific or tulpamancer-specific neuroimaging, stimulation, or longitudinal training evidence establishing a mechanism. That absence is the opportunity, and most of this paper is about the shape of the experiment that would fill it.

The awkward result, and the model that predicts it

A naive account says members of nonhuman-identifying communities have generally more plastic body models and should therefore show stronger illusions. One direct study exists. It found the opposite. Fifty furies reported weaker subjective embodiment of a human rubber hand than comparison data, and within the sample, participants who valued the human form less showed less proprioceptive drift toward the human hand (Kranjec et al., 2019). The study compares against previously collected data rather than a contemporaneous control group and appears in conference proceedings, and both limitations should be carried wherever it is used. Taken at face value, it refutes the account that motivated the question.

It is also exactly what a morphology-congruent account predicts, and the paper's second computation shows this rather than asserting it. Body ownership can be modeled as inference about whether visual and proprioceptive evidence share a common bodily cause (Samad, Chung and Shams, 2015). Add a prior over morphological correspondence, so that how readily an effector is taken as part of the body depends on how well its form matches the participant's self-model. Shift that single prior for an identifying group, lowering the prior mass on a human effector and raising it on a congruent nonhuman one, and the model produces weaker ownership of the human hand by 0.22 and stronger ownership of the congruent nonhuman body by 0.22, with a neutral nonhuman morphology unmoved (Figure 1, left). One knob, two signs. The field's most awkward finding becomes its cleanest support, and the prediction that comes with it is testable: the same participants who bind a human hand less should bind a congruent nonhuman body more.

The same model reproduces the cat-ear dissociation as a regime rather than a puzzle. When ownership is read as the posterior probability of a common cause and drift as the resulting shift in felt position, a manipulation that changes the prior moves both, but not in fixed proportion: at small visual-proprioceptive discrepancy the ownership change dominates and drift barely moves, while at large discrepancy drift dominates (Figure 1, right). Ownership without drift is therefore

expected in a specific and identifiable part of the parameter space, which turns a null result into a positioned measurement.

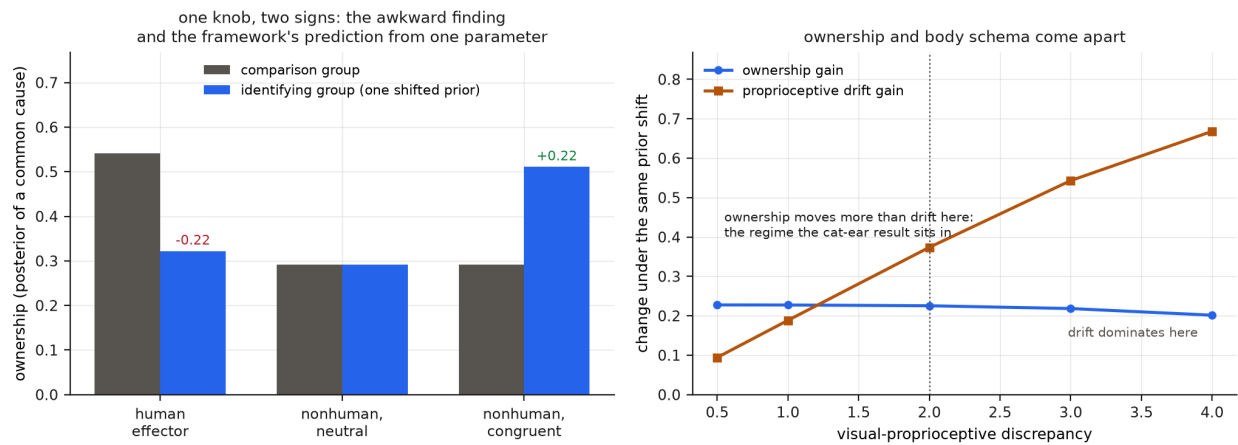


Figure 1: Morphology-congruent binding. Left: a single shifted prior over morphological correspondence lowers ownership of a human effector and raises it for a congruent nonhuman one by the same amount, reproducing the observed sign and predicting its mirror. Right: under the same shift, ownership and proprioceptive drift change in different proportions depending on the sensory discrepancy, so ownership without drift is a regime rather than an anomaly.

What the decisive experiment costs

Three theories are live. Global plasticity says identifying participants embody everything more readily. Morphology-congruent binding says they bind bodies matching their self-model and human bodies less. Expectancy says the differences live in subjective report and track hypothesis awareness, which is a serious possibility given that rubber-hand reports respond to demand characteristics and that ownership and drift dissociate. The paper's first computation asks a design question: under which experiment can these be told apart?

Data were simulated from each theory in turn and all three fit back by model selection, under the design the literature has run and under richer alternatives (Figure 2). With a single human effector, recovery is 0.51 against a chance level of 0.33, and the specific failure matters more than the average: when morphology-congruent binding is the truth, that design selects global plasticity 96 percent of the time. The field's one direct study is built exactly this way, which means it could not have distinguished the account it appears to refute from the account it actually supports. Adding a generic nonhuman avatar raises recovery to 0.73 and identifies congruent binding only 41 percent of the time, because a neutral nonhuman body is precisely the cell where the two theories agree. Only the crossed design, varying morphology against identity congruence within participants, reaches 0.94 and recovers congruent binding 97 percent of the time.

The cost follows. Because the diagnostic quantity is an interaction rather than a main effect, the sample must be sized for the interaction: about 30 participants per group for 80 percent power at the modeled effect, against the 20 or so typical of embodiment experiments, and more once

multiple groups and multiple outcome families enter. That number is the paper’s most practical output. “Nobody has run the right experiment” becomes a protocol with a design matrix and a sample size attached, and the protocol’s decisive test is an identity-congruent interaction appearing in at least one subjective measure and at least one behavioral, physiological, or psychophysical measure, since a questionnaire difference alone would not distinguish a changed body model from a changed expectation.

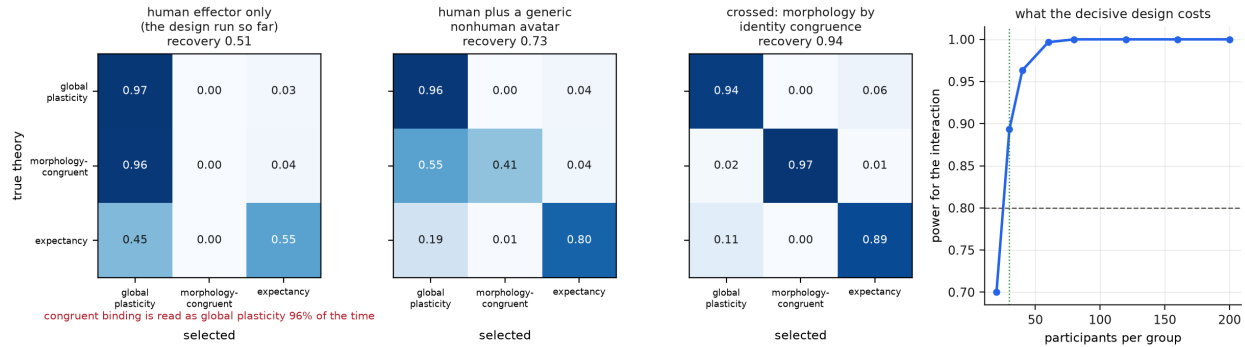


Figure 2: Design discrimination. The three matrices show which theory model selection picks when each is true, under the design run so far, under a generic-nonhuman variant, and under the crossed design; the human-only design reads morphology-congruent binding as global plasticity 96 percent of the time. Right: power for the diagnostic interaction against sample size.

Fast states, slow models, and what survives

A rubber-hand illusion takes seconds and a fursona takes years, so no single timescale holds both. The formalism separates three things: a fast neural and physiological state driven by perturbation, stable individual traits, and slowly learned generative-model parameters that accumulate from the fast state and decay when it stops. The safely reachable phenotype is then the set of configurations attainable under bounded, ethically permissible perturbations within a horizon, which is a reachable set in the control-theoretic sense with a hard safety constraint rather than a soft penalty.

Three quantities become measurable in this frame. Hysteresis is the fraction of a trained configuration that survives after the driving perturbation stops, computed here as 55 percent of the trained gain. It separates a learned identity from a maintained illusion. An illusion collapses with its stimulus; a learned model does not. The learning rates for binding a representation to SELF and to OTHER are separately estimable, which makes the paper’s central hypothesis quantitative. And the reachable region under training exceeds the fast-driven baseline by a factor of 2.7 in the model, which is the formal statement of why longitudinal designs are worth their cost: most of the interesting territory is unreachable in one session.

The generative-binding hypothesis now has an informative form. Therian-like identification may involve morphology-congruent binding of nonhuman representations to SELF; cultivated inner agents may involve learned binding of internally generated representations to OTHER. The prediction is a crossed sign under matched training with a novel creature that no participant has prior

attachment to: faster SELF binding in one group, faster OTHER autonomization in another, with the comparison group at neither extreme (Figure 3, left). A main effect would not do. General suggestibility produces one of those. Under the modeled learning-rate gap the crossing is detectable at about 15 participants per group, which is encouraging, and it is the one prediction here that a modest study could kill.

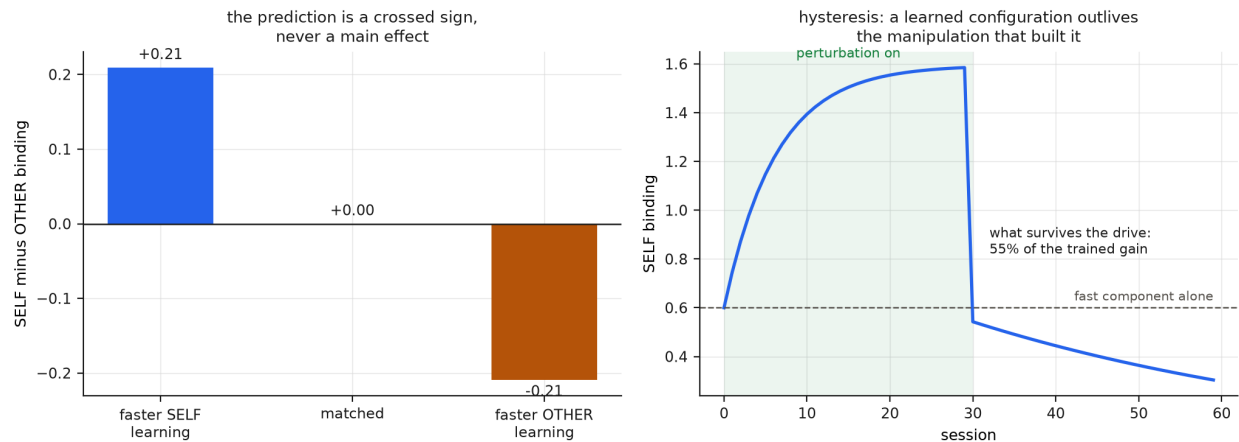


Figure 3: The multiscale reachable set. Left: the SELF and OTHER hypothesis predicts a crossed sign across profiles rather than a main effect, which is what distinguishes it from general suggestibility. Right: hysteresis, with 55 percent of the trained gain surviving after the perturbation stops, which is what separates a learned configuration from a maintained illusion.

Retiring the metaphor that started this

The intuition behind such enquiries is often that the self is software and that unusual identities are a configuration loaded wrong. The metaphor gets one thing right. The adult self is not simply read off the biology, and some process determines what counts as body, self, and agent. It gets four things wrong, and naming them is part of the paper's work. It implies a single initialization, where the evidence indicates continuous recalibration from sensorimotor history, interoception, social feedback, and current state. It implies a discrete species template, which cannot account for identities whose referents are mythological or invented rather than zoological, and which are better located at semantic and autobiographical levels than in any morphology library. It conflates bodily ownership with narrative identity, when a person can enjoy an avatar, feel represented by it, report phantom sensations, and describe themselves in its terms without those being one psychological operation. And it makes difference equivalent to malfunction, when a persistent nonstandard self-model may be neutral, valuable, or distressing, and function has to be measured rather than inferred from unusual content. The better vocabulary is morphology-congruent binding, agent-model autonomization, hysteresis, and reachable-set mapping. Less evocative, considerably more testable.

Natural experiments, held honestly

Populations who already show unusual transitions are useful starting coordinates, and the fever literature is the paper's model of how to hold one honestly. An early prospective study of 30 febrile and 30 afebrile autistic children found transient reductions on several aberrant-behavior measures during fever (Curran et al., 2007). A later prospective study following 141 autistic children found that of the 50 who had a fever, 3 showed reported improvement, with the typical pattern being worsening (Byrne et al., 2022). The honest summary carries both: fever-associated improvement may offer a rare reversible natural experiment in state-dependent behavioral access, and it is uncommon, and the phenomenon does not license the claim that fever unlocks dormant capacities. The reversible structure, from baseline to altered state to recovery, is what makes any such case valuable, because reversibility is what distinguishes a state transition from a stable trait difference.

Ethics, communities, and dual use

The people this paper discusses are not a natural resource for cognitive science. Three commitments follow. First, terminology belongs to communities, and the paper uses their terms as they use them; cognitive interviews should precede any measurement, so that a spiritual identity, a role, a self-representation, a bodily sensation, and a literal biological belief are not collapsed into one variable called animal identification. Second, difference is not disorder: the clinical literature enters only as a boundary condition and with its authors' own warning attached, distress and impairment are measured rather than assumed, and no finding about a group licenses inference about an individual. Third, participants recruited from small online communities are re-identifiable in ways general-population samples are not. That constrains data sharing and case description. Four further commitments: preregistered primary outcomes, expectancy measured rather than hoped away, community consultation without giving any party control over conclusions, and null results published. The closed-loop exploration this framework invites, where an algorithm proposes the next perturbation to maximize expected information about the phenotype, needs the safety constraint to be hard rather than a term in an objective, and any novel state it reports needs the full sequence from within-person replication through independent replication before it is called a discovery rather than a measurement artifact.

Objections

The strongest objection is that the simulations contain no people, so the numbers are properties of assumptions. That is correct, stated wherever a number appears, and it is what design analysis is: the recovery matrices and the sample size are claims about what a given experiment could resolve under a stated effect, and they are the right kind of claim to make before spending participants rather than after. The second objection is that a framework spanning fever, avatars, and identity is too broad to constrain anything. At the mechanistic level it is. These phenomena share a method: repeated within-person measurement under reversible perturbation, with factorized outcomes. They probably share no mechanism. Fever involves immune and metabolic state; embodiment involves multisensory inference; cultivated agents involve learned source attribution. A

third objection is that the enterprise risks pathologizing communities by studying them as unusual cases. The risk is real. The framework's logic runs the other way: these are populations whose configurations demonstrate what the ordinary nervous system can do, and the null result the design analysis most protects against is a group difference in questionnaires being read as a difference in bodily computation. A fourth objection: the one direct study is a proceedings paper with historical controls, so building a framework on it overreads a thin result. Agreed. The paper's use of it is limited to a sign that a model reproduces, with the recommendation that the study be repeated with contemporaneous controls in the crossed design specified here.

Falsification

The framework fails if the phenotype coordinates collapse: if ownership, body schema, agency, identity, and reality testing load on one factor across manipulations, the factorization was decoration, and the cat-ear dissociation would have to be a measurement failure rather than a result. The morphology-congruent account fails if group differences appear equally across every morphology, which is the global-plasticity signature the crossed design is built to detect. The SELF and OTHER hypothesis fails if both communities show the same undifferentiated effect, since a shared main effect is what suggestibility predicts. The learned-autonomization account fails if longitudinal training produces no change beyond practice effects, and the hysteresis claim fails if trained configurations vanish the moment the manipulation stops. The trait account fails in the other direction if group differences vanish after adjusting for expectancy and imagery, which would relocate the phenomenon from body computation to report. And the paper's design analysis itself fails if a well-powered crossed experiment recovers the theories no better than the human-only design did in simulation, which would mean the generative models here misdescribe where the information lives.

What is actually being proposed

The claim is modest at the level of mechanism and immodest at the level of scope. No mechanism is established here. No community is characterized, and no capability is unlocked. What is proposed is that capability, embodiment, agency, and identity are coordinates of one measurable object; that the object is state-dependent and only partly explored; that the phenomena which look strangest from outside are among its more informative regions; and that the experiment which would settle the current question is crossed and costs about 30 participants per group rather than the 20 the field is used to. The last of those is the part someone can act on tomorrow. A framework that cannot say what to measure next is a vocabulary, and the difference between the two is a sample size.

Reproducibility

The simulation, its 3 figures, and every model number quoted above regenerate from one command in the paper's repository (`simulation/run_all.py`; seeded, bit for bit). 20 invariants, among them the human-only design's failure to discriminate and its specific misreading of congruent binding, the crossed design's recovery, the monotone power curve and its threshold, the one-knob-two-

signs property with its neutral-morphology control, the ownership-over-drift regime at small discrepancy and its reversal at large, the positive but incomplete hysteresis, the crossed rather than main-effect form of the double dissociation, and the trained region exceeding the fast-driven baseline, fail the run loudly if broken. Every quantity is a property of the stated generative models and none is an empirical estimate; the externally measured figures quoted in prose enter results.json as cited records with their sources.

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